

Discussion of "Optimal Monetary Policy in HANK Economies"

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Discussed by
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Disclaimer: The views expressed do not necessarily reflect the views of the Bank of Spain or the Euro-System

Model and main results

"Relatively simple" HANK-type environment

- **CARA** utility + **gaussian** idiosyncratic earnings shocks
- **Incomplete** markets and a **natural borrowing limit**
- **NK block** generates a standard Phillips curve
- Standard monetary-fiscal authority controls **nominal interest rates** and **labour tax**.
- **Utilitarian** social welfare function

Main results

- 1 **Analytically tractable** Ramsey problem, with **consumption dispersion (CD)** as a key component
- 2 Incentive to decrease CD implies **optimal deviation** from production efficiency and 0-inflation
- 3 New **time-inconsistency** in optimal policy.
- 4 Features of such deviation depend crucially on **cyclical** of income risk

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Key features & mechanism

- 1 **TRACTABILITY:** CARA + gaussian shocks + no borrowing limit imply
 - Precautionary savings ($u''' > 0$)
 - Linear consumption / saving rules
 - Homogeneous MPC which is time-varying
 - Linear Aggregation
- 2 **CONSUMPTION DISPERSION:** given incomplete markets, CD driven by two key forces:
 - amount of income risk
 - sensitivity of consumption to income changes
- 3 **OPTIMAL POLICY:** production efficiency + inequality reduction:
 - pro-cyclical risk: trade-off between reduction of risk and reduction of pass-through implies deviation from RANK gets dampened
 - counter-cyclical risk: more persistent departures from "divine coincidence"

Optimal monetary policy in HANK: the contestants

Mechanisms					Optimal MP	
NK	Fisher	UIE	Earn. Heterog.		Time-0	Time- ∞

**Acharya
et al. (2019)**

Bhandari
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Nuño
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RANK

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RANK	✓	X	X	X		

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Comment I: Time-0 vs. Time- ∞ in the basic RANK

In the basic textbook RANK (see Gali (2007) ch. 5 or Woodford (2003) ch. 6-7) the **Ramsey problem under commitment** is to choose a path $\{\pi_t, y_t\}$, $t \geq 0$ that maximizes

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \left\{ \underbrace{\frac{1}{2} \left[\pi_t^2 + \lambda (x_t - x^*)^2 \right]}_{\text{period-welfare}} + \underbrace{\psi_t [\pi_t - \kappa x_t - \beta \pi_{t+1}]}_{\text{period-Phillips curve}} \right\} \quad (4.1)$$

with ψ_t the lagrange multiplier associated with the period-t PC. The FOCs are similar to the ones in Acharya et al. (2019)

$$\pi_t + \psi_t - \psi_{t-1} = 0 \quad (4.2)$$

$$\lambda (x_t - x^*) - \kappa \psi_t = 0 \quad (4.3)$$

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- **Time-0 optimal policy** : optimal policy chosen taking as given expectations at $t = -1$ and the state at $t = 0$.

$$\psi_{-1} = 0 \quad (4.4)$$

"Doing something now, but promising to behave differently in the future"

⇒ Inflationary bias at $t=0$

⇒ Time-inconsistency of optimal policy

- **Time-less optimal policy**: optimal policy chosen such that evolution at $t = 0$ is consistent with what was expected from such policy in the past.

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Comment II: Cyclical Income Risk

- Cyclical property of **income risk** key for determinacy / stability and dynamic properties of model.
- The theoretical results rely on income broadly defined, including **labour** and **non-labour** sources.
- This income cyclical property depends on the parameter

$$\phi = \frac{\partial \text{Var}(x)}{\partial y} \equiv \text{cyclical property of cash on hand} \quad (4.6)$$

- Guvenen et al. (2014) results (**countercyclical left skewness**) are only about **labour** income: W2 form in the US Social Security Administration records. This includes **wages and salaries, bonuses and exercised stock options**
- We know less (empirically) about the cyclical properties of income...

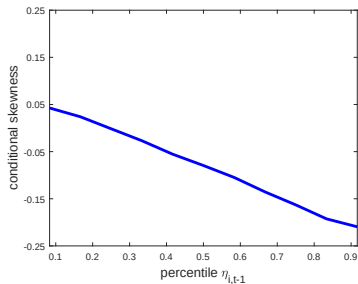
Quick estimation using Spanish household data

- I estimate a version of [Arellano et al. \(2017\)](#) labour income process using household data for Spain 2002-2014.
- For a given household i of age j at time t , persistent income $y_{i,j,t}$ is assumed to follow a [generalized](#) Markov process.
- In particular, the τ^{th} conditional quantile of persistent income given its past $y_{i,j-1,t-1}$ is assumed to be:

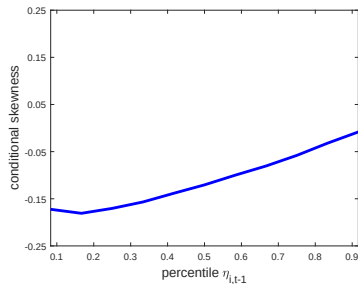
$$y_{i,j,t} = Q_{j,t}(y_{i,j-1,t-1}, u_{i,j,t}) \quad , \quad u_{i,j,t} | y_{i,j-1,t-1} \dots \sim U[0, 1]$$

- In the data, $y_{i,j,t}$ is defined as the [residual](#) from a regression of log-labour income on a set of covariates including age of the household head, education categories, family size, etc.

Conditional Skewness - Average age



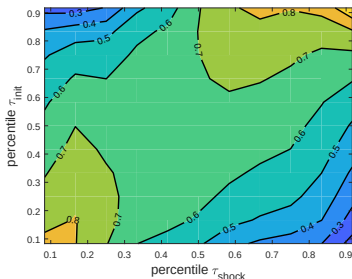
2002-2008



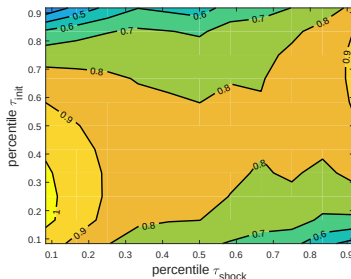
2008-2014

Persistence of earnings histories - Average age

$$\rho(\nu_{i,t-1}\tau_{i,t}) = \frac{\partial Q_{\nu|\nu-1}(\nu_{i,t-1}, \tau_{i,t})}{\partial \nu} \quad (4.7)$$



2002-2008



2008-2014

Some unfair comments: Size and homogeneity of MPC

- My impression is that with current parametrization (even with $r = 4\%$!), the homogeneous MPC μ is quite small.
- MP affects μ through easiness to borrow and save to smooth shocks.
- Data: between wealthy and poor HTM, almost 40%-50% of households in the UK-US ...
- ... how meaningful is the self insurance channel?

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MPCs:

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- ... how meaningful is the self insurance channel?

Welfare: dynamics under different shocks?

Life-cycle: what is the role of the perpetual-youth structure in OMP?

Overall

- A very promising and interesting paper!
- Fair enough, I have a personal bias to like it...
- ... but together with the PRANK paper by Acharya & Dogra (2018), I think they form a combo which sheds A LOT of light on the current RANK-TANK-HANK-SAM literature.

HANK

THANKs!